

The Wideband Delphi Process

• Step 1: Choose the team

- The project manager selects the estimation team and a moderator. The team should consist of 3 to 7 project team members.
 - The moderator should be familiar with the Delphi process, but should not have a stake in the outcome of the session if possible.
 - If possible, the project manager should not be the moderator because he should ideally be part of the estimation team.

The Wideband Delphi Process

• Step 2: Kickoff Meeting

- The project manager must make sure that each team member understands the Delphi process, has read the vision and scope document and any other documentation, and is familiar with the project background and needs.
- The team brainstorms and writes down assumptions.
- The team has the generated WBS.
- The team agrees on a unit of estimation.

• Step 3: Individual Preparation

- Each team member independently generates a set of preparation results.
- For each task, the team member writes down an estimate for the effort required to complete the task,
- and any additional assumptions he needed to make in order to generate the estimate.

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- Step 4: Estimation Session

- During the estimation session, the team comes to a consensus on the effort required for each task in the WBS.
- Each team member fills out an estimation form which contains his estimates.
- The rest of the estimation session is divided into rounds during which each estimation team member revises her estimates based on a group discussion. Individual numbers are not discussed.

The Wideband Delphi Process

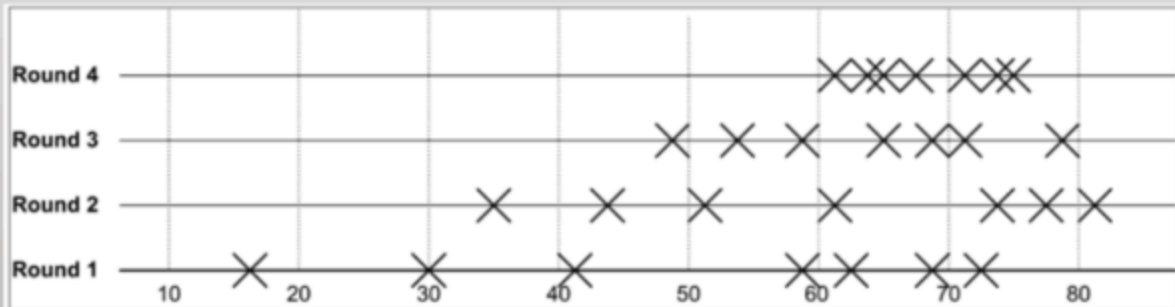
- Step 4: **Estimation Session** (continued)

- The moderator collects the estimation forms and plots the sum of the effort from each form on a line:



- Step 4: Estimation Session (continued)

- The team resolves any issues or disagreements that are brought up.
 - Individual estimate times are not discussed. These disagreements are usually about the tasks themselves. Disagreements are often resolved by adding assumptions.
- The estimators all revise their individual estimates. The moderator updates the plot with the new total:
- Goal: to achieve the max Convergence.



- Step 4: Estimation Session (continued): /dELTA

- The moderator leads the team through several rounds of estimates to gain consensus on the estimates. The estimation session continues until the estimates converge or the team is unwilling to revise estimates.

- Step 5: Assemble Tasks

- The project manager works with the team to collect the estimates from the team members at the end of the meeting and compiles the final task list, estimates and assumptions.

- Step 6: Review Results

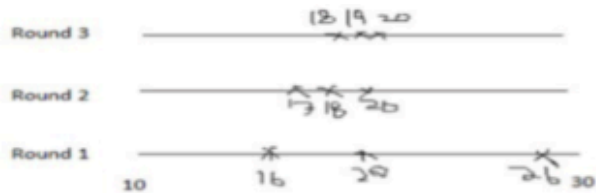
- The project manager reviews the final task list with the estimation team.

For creating a Chat Application, A list of activities having different tasks are formulated and on the basis of that, project estimation using Wide Band Delphi technique is done. Three estimators took part in the estimation process, shown in Chart given below. Below are the estimation forms deduced from the kickoff meetings. The numbers in the tables depicts the effort estimation in terms of days required for each activity.

Member: Salas	Round 1	Round 2	Round 3	Member: Aliya	Round 1	Round 2	Round 3	Member: Ahmed	Round 1	Round 2	Round 3
Create UI	3	3	3	Create UI	4	4	4	Create UI	7	7	7
Create groups	3	3	4	Create groups	6	5	5	Create groups	8	3	3
Send message	6	6	6	Send message	2	5	5	Send message	7	6	5
Receive message	3	3	3	Receive message	3	3	3	Receive message	2	2	3
Add member in groups	5	2	2	Add member in groups	1	1	2	Add member in groups	5	2	2

For the above data, do the following:

- Show how the meeting progressed by drawing a single result graph that represents cumulative estimation for the three members.
- Create the effort estimation Sheet showing all best, worst and average cases for Round 1 only.



b)

	Salas	Aliya	Ahmed	Best	Worst	Average
Create UI	3	4	7	3	7	5
Create groups	3	6	8	3	8	6
Send message	6	2	7	2	7	5
Receive message	3	3	2	2	3	3
Add member in group	5	1	5	1	5	3
	20	16	29			

^ effort estimation sheet

Step 1: Find total estimate of each member for every round

Add all activity estimates of each member.

Round 1

Salas

- Create UI = 3
- Create groups = 3
- Send message = 6
- Receive message = 3
- Add member = 5

$$\text{Total} = 3 + 3 + 6 + 3 + 5 = 20$$

Aliya

- $4 + 6 + 2 + 3 + 1 = 16$

Ahmed

- $7 + 8 + 7 + 2 + 5 = 29$

So Round 1 points on graph:

- Salas = 20
- Aliya = 16
- Ahmed = 29

Best = 16 (minimum)

Worst = 29 (maximum)

Round 2

Salas

- $3 + 3 + 6 + 3 + 2 = 17$

Aliya

- $4 + 5 + 5 + 3 + 1 = 18$

Ahmed

- $7 + 3 + 6 + 2 + 2 = 20$

Round 2 points:

- 17, 18, 20

Best = 17

Worst = 20

Round 3

Salas

- $3 + 4 + 6 + 3 + 2 = 18$

Aliya

- $4 + 5 + 5 + 3 + 2 = 19$

Ahmed

- $7 + 3 + 5 + 3 + 2 = 20$

Round 3 points:

- 18, 19, 20

Best = 18

Worst = 20

Notice how estimates are converging:

Round 1 spread = $29 - 16 = 13$

Round 2 spread = $20 - 17 = 3$

Round 3 spread = $20 - 18 = 2$

This shows consensus being reached.

Part (a) Result Graph

Plot only the total estimates:

Round	Salas	Aliya	Ahmed	Best	Worst
1	20	16	29	16	29
2	17	18	20	17	20
3	18	19	20	18	20

The graph in your solution is showing these ranges:

- Round 1 → 16 to 29
- Round 2 → 17 to 20
- Round 3 → 18 to 20

Part (b) Best, Worst, Average for Round 1 Activities

Take estimates of all three members for each activity.

Create UI

Values = 3, 4, 7

Best = min = 3

Worst = max = 7

Average:

$$\frac{3 + 4 + 7}{3} = \frac{14}{3} = 4.67$$

≈ 5